# Practice workbook · Identify bones and bone tissues on a diagram

Naming bones of the skeleton and the tissues of a long bone from drawings or images.

8 question parts: 1 worked-example retry and 7 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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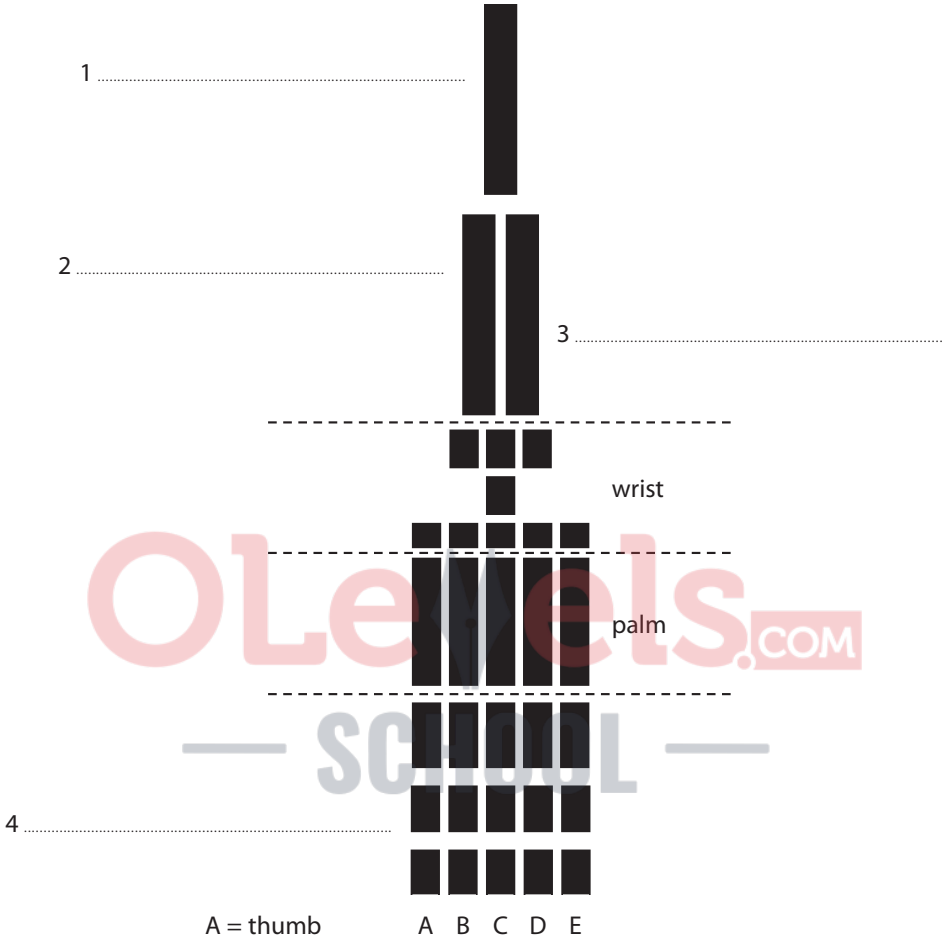
Type ID: HBI-SKELETON-BONE-IDENTIFY

# 1. Question 2 · part a.i · 4 marks

June 2021 Paper 1

Worked-example retry · Attempt part a.i only.

**2** The diagram shows a model representing the arrangement of bones in the arm.




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(a) (i) Name the bones labelled 1, 2, 3 and 4 on the diagram.

**(4)**

(ii) A hinge joint is a synovial joint.

Draw a circle on the diagram to show where a hinge joint is found.

**(1)**

(iii) Name two other types of synovial joint found in the skeleton.

**(2)**

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2 .....

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## 1. Question 2 · part a.i · 4 marks · Working and answer

Working and answer

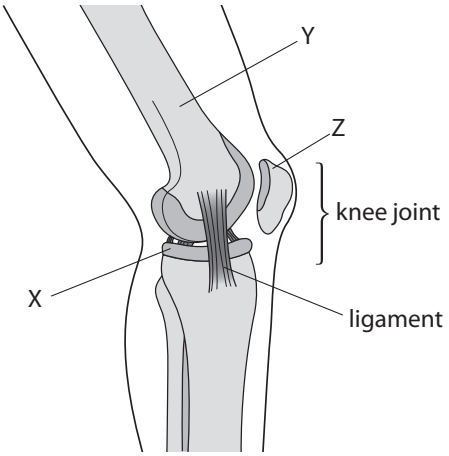
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## 2. Question 1 · part a.i · 3 marks

June 2019 Paper 1

Further practice · Attempt part a.i only.

**1** (a) The diagram shows structures in part of a human leg.




(i) The box lists structures in the human leg.

| | | | |
|---|---|---|---|
| cartilage | femur | kneecap | muscle |
| fibula | tendon | tibia | |

Use words from the box to name structures X, Y, and Z.

**(3)**

X .....

Y .....

Z .....

(ii) What is the function of ligaments in the knee joint?

**(1)**

- ☒ **A** ligaments attach muscles to the leg bones
- ☒ **B** ligaments hold the leg bones together
- ☒ **C** ligaments move the leg bones
- ☒ **D** ligaments stop the leg bones rubbing together

(iii) The type of joint found in the knee and the elbow is the same.

Give the name of this type of joint.

**(1)**

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## 2. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

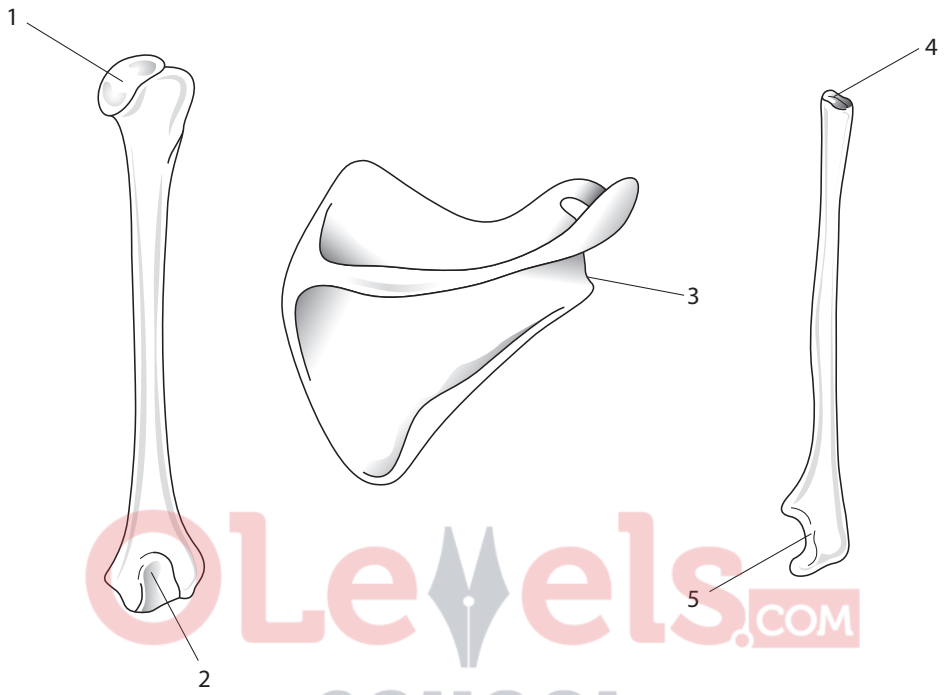
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# 3. Question 2 · part b.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.i only.

**2** (a) The diagrams show three bones from the human skeleton.




Which two numbered regions come together to form a ball and socket joint?

**(1)**

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

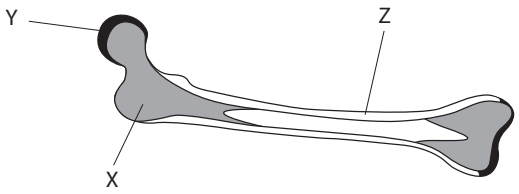
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# 3. Question 2 · part b.i · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.i only.

(b) The diagram shows a section through a bone.




(i) Identify the part labelled X.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

(1)

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

(1)

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(iv) Describe the function of the part labelled Y.

(2)

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## 3. Question 2 · part b.i · 1 mark · Working and answer

Working and answer

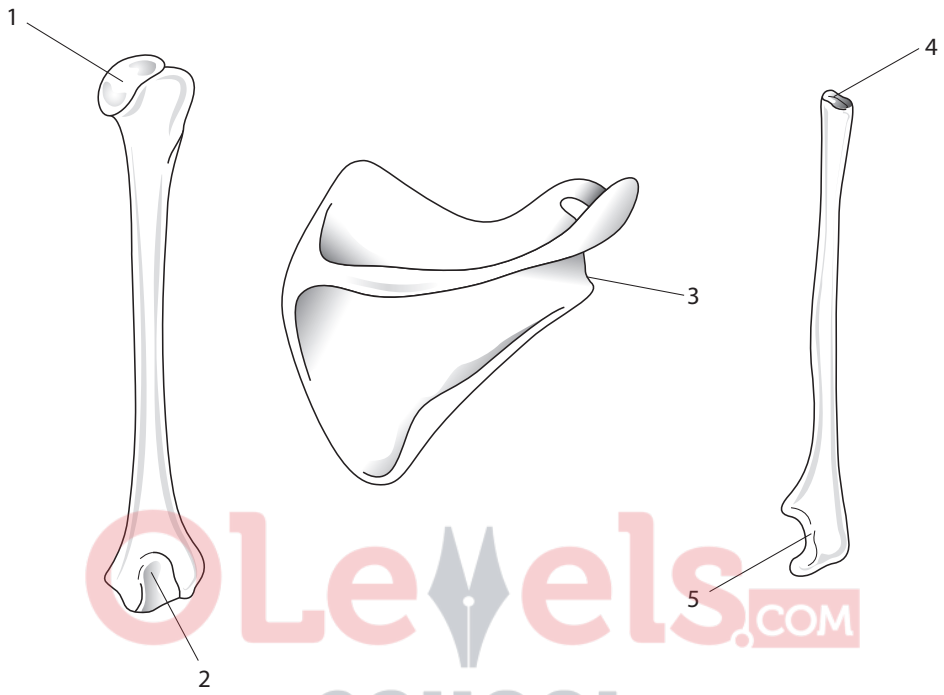
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# 4. Question 2 · part b.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

**2** (a) The diagrams show three bones from the human skeleton.




Which two numbered regions come together to form a ball and socket joint?

**(1)**

- ☒ **A** 1 and 3
- ☒ **B** 1 and 5
- ☒ **C** 2 and 3
- ☒ **D** 2 and 4

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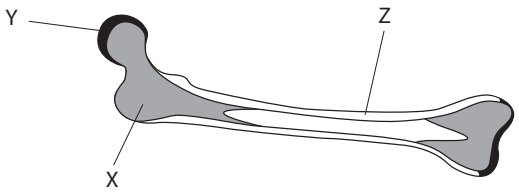
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# 4. Question 2 · part b.ii · 1 mark

June 2019 Paper 2R

Further practice · Attempt part b.ii only.

(b) The diagram shows a section through a bone.




(i) Identify the part labelled X.

**(1)**

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(ii) Identify the part labelled Z.

**(1)**

- ☒ **A** cartilage
- ☒ **B** compact bone
- ☒ **C** ligament
- ☒ **D** spongy bone

(iii) Name the part labelled Y.

**(1)**

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(iv) Describe the function of the part labelled Y.

**(2)**

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## 4. Question 2 · part b.ii · 1 mark · Working and answer

Working and answer

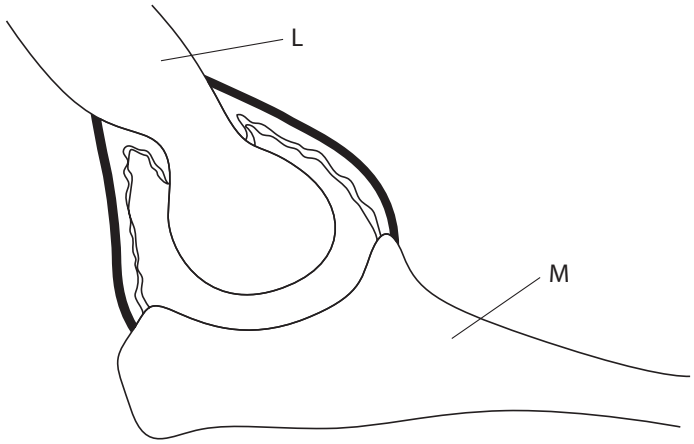
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## 5. Question 6 · part b.i · 2 marks

November 2021 Paper 2

Further practice · Attempt part b.i only.

**6** The diagram shows some of the structures of the elbow joint.




(a) State what is meant by the term **joint**.

**(2)**

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(b) (i) Give the name of bone L and the name of bone M.

**(2)**

L .....

M .....

(ii) Cartilage has not been shown on the diagram.

Draw the missing cartilage on the diagram.

**(2)**

**20**

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## 5. Question 6 · part b.i · 2 marks · Working and answer

Working and answer

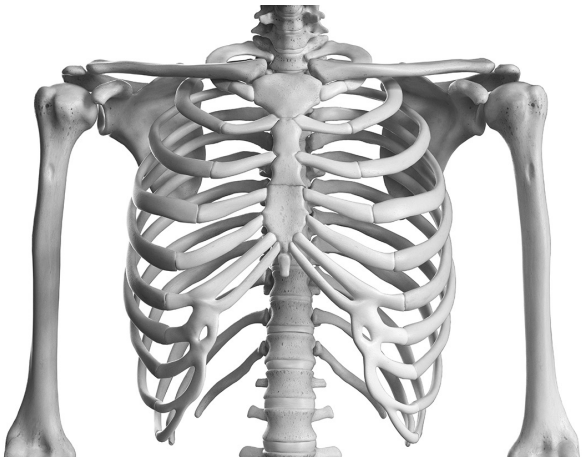
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# 6. Question 1 · part a.i · 3 marks

January 2023 Paper 2

Further practice · Attempt part a.i only.

**1** (a) The image shows part of the skeleton.



(Source: © SciePro/Shutterstock)

(i) Label the clavicle, scapula and vertebral column on the image. **(3)**

(ii) The vertebral column is part of the axial skeleton.

Describe the functions of the axial skeleton. **(2)**

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## 6. Question 1 · part a.i · 3 marks · Working and answer

Working and answer

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# 7. Question 2 · part b.i · 2 marks

June 2023 Paper 1R

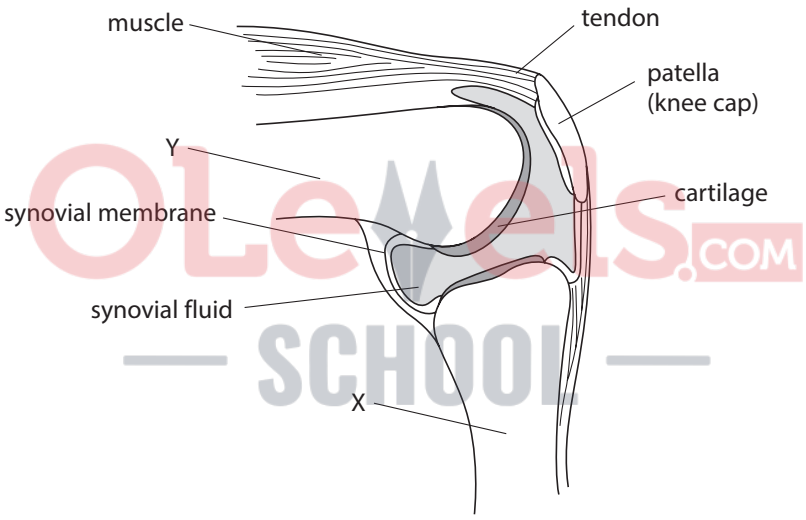
Further practice · Attempt part b.i only.

**2** The skeleton contains joints.

(a) Describe what is meant by the term **joint**.

**(2)**

(b) The diagram shows a knee joint.




(i) Identify bone X and bone Y.

**(2)**

X .....

Y .....

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## 7. Question 2 · part b.i · 2 marks

June 2023 Paper 1R

Further practice · Attempt part b.i only.

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(ii) Give the functions of tendons and cartilage.

(4)

tendons

cartilage

(iii) Give two structures, not shown on the diagram, that would be necessary for the joint to fully function.

(2)

1 .....

2 .....

**(Total for Question 2 = 10 marks)**

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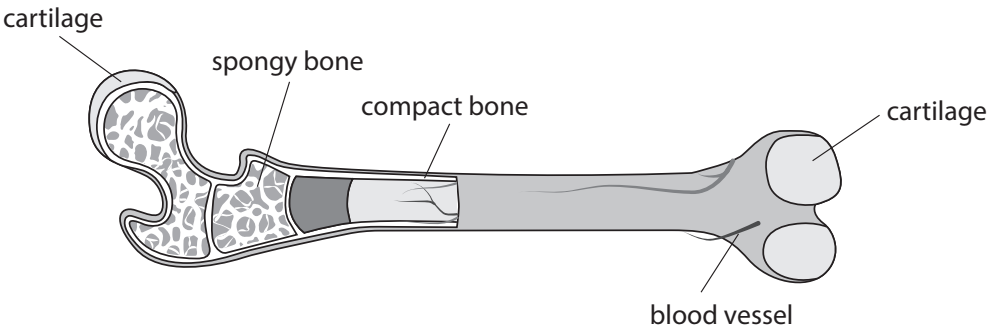
Working and answer

# 8. Question 1 · part a.i · 1 mark

June 2025 Paper 2R

Further practice · Attempt part a.i only.

**1** (a) The diagram shows a bone found in the human body.




(Source: adapted from: © Designua / Shutterstock)

(i) State the name of the type of bone shown in the diagram.

**(1)**

.....

(ii) Place one tick (✓) in each row to give the function of each part of the bone.

**(3)**

| Part | Function | | |
|---|---|---|---|
| | reduces friction | where red blood cells are produced | provides strength |
| compact bone | | | |
| spongy bone | | | |
| cartilage | | | |

(iii) Describe the function of the blood vessel in the bone.

**(2)**

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## 8. Question 1 · part a.i · 1 mark · Working and answer

Working and answer

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